

# BCS Gap Equation in SCm Vacuum

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## PAPER\_949: BCS Gap Equation in SCm Vacuum

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**Session:** 214  
**Source:** bcs\_superconductivity\_uqff.py (BCSGapEquation)  
**Calculator:** BCSGapEquationCalc (CP4 #533)  
**CVW:** v2.0.0 compliant

### Abstract

We derive the BCS energy gap equation in the SCm vacuum phonon framework. The gap  $\Delta$  is determined self-consistently via  $\Delta = (\hbar \omega_{\text{SCm}}/2) \cdot \tanh(\Delta/2k_{\text{BT}}) \cdot S_{\{26\}} \cdot (F_{\text{U},\text{Bi}}/F_{\text{U}})$ , where  $\omega_{\text{SCm}} = 2\pi \cdot 1.25 \text{ THz}$  is the SCm phonon resonance frequency. This maps conventional BCS superconductivity to the UQFF vacuum structure, with the 26-layer buoyancy sum  $S_{\{26\}}$  replacing the Debye phonon spectrum.

### 1. Gap Equation

$$\Delta = \frac{\hbar \omega_{\text{SCm}}}{2} \tanh\left(\frac{\Delta}{2k_{\text{BT}}}\right) \cdot S_{\{26\}} \cdot \frac{F_{\text{U},\text{Bi}}}{F_{\text{U}}}$$

Self-consistent solution via iterative fixed-point method converges in  $<50$  iterations at all temperatures.

### 2. Temperature Dependence

| $T \text{ (K)}$ | $\Delta \text{ (eV)}$ | $\Delta/\Delta_0$ |
|-----------------|-----------------------|-------------------|
| 0               | $\Delta_0$            | 1.000             |
| 1               | $\approx \Delta_0$    | $\approx 1.000$   |
| 100             | reduced               | $< 1$             |
| $T_{\text{c}}$  | 0                     | 0                 |

### 3. Source Data

- **File:** bcs\_superconductivity\_uqff.py
- **Session:** 214
- **CP4 Class:** BCSGapEquationCalc (#533)

## References

1. Murphy, D.T. --- Star Magic UQFF Framework (2024-2026)
2. Bardeen, J., Cooper, L.N. & Schrieffer, J.R. (1957) --- Phys. Rev. 108, 1175
3. PAPER\_950 --- BCS Critical Temperature
4. PAPER\_951 --- Cooper Pair Phonon Coupling
5. PAPER\_955 --- BCS Phonon Resonance
6. PAPER\_957 --- Cooper Pair Lagrangian Variation
7. PAPER\_877 --- Cosmogenesis Master Lagrangian

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## Cross-Links

| Related Paper | Relationship |  
|-----|-----|  
| PAPER\_950 |  $T_c$  derived from this gap equation |  
| PAPER\_951 | Cooper pair coupling via  $V_{\text{eff}}$  |  
| PAPER\_952 | Spectral ladder provides phonon channel hierarchy |  
| PAPER\_955 | BCS gap at  $\omega_{\text{SCm}}$  resonance |  
| PAPER\_957 | Lagrangian variational principle yields this gap |

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<!-- PKG-LAG-S225 -->

## Session 225 Phonon-Physics Upgrade: UQFF 9-Sector Lagrangian

> Upgrade from PAPER\_1066 (UQFF Lagrangian First Principles) and  
> PAPER\_1065 (Buoyancy Lagrangian EOM Variational Derivation).

The complete UQFF Lagrangian density, from which all sector-specific equations of motion derive:

$$\mathcal{L}_{\text{UQFF}} = \mathcal{L}_{\text{GR}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{phonon}} + \mathcal{L}_{\text{interaction}} \quad \mathcal{L}_{\text{phonon}}$$

$$\mathcal{L}_{\text{SCm}} = \frac{1}{2}(\partial_\mu \phi)^2 - \lambda |\phi|^2 - v_{\text{SCm}}^2 |\phi|^2$$

The SCm condensate potential minimum gives  $V(\phi_0) = -7.09 \times 10^{-37} \text{ J/m}^3$  (matching  $\rho_{\text{SCm}}$ ) and phonon mass  $m_{\text{phonon}} = \sqrt{8\lambda} v_{\text{SCm}}$ .

### Nine-sector closure (Session 202):

$$\mathcal{L}_9 = \mathcal{L}_{\text{EH}} + \mathcal{L}_{\text{YM}} + \mathcal{L}_{\text{Dirac}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{mag}} + \mathcal{L}_{\text{buoy}} + \mathcal{L}_{\text{aether}} + \mathcal{L}_{\text{LENR}} + \mathcal{L}_{\text{KK}}$$

| Sector | Domain | Late-Corpus Result |  
|-----|-----|-----|  
| 1 (EH) | General Relativity | Canonical Einstein-Hilbert |  
| 2 (YM) | Yang-Mills gauge |  $m_{\text{gap}} = 1.736 \text{ GeV}$  (PAPER\_1318) |  
| 3 (Dirac) | Fermion / LENR | Kozima neutron-drop (PAPER\_1061) |  
| 4 (SCm) | Superconducting manifold |  $V(\phi_0) = -\rho_{\text{SCm}}$  canonical |  
| 5 (Mag) | Um magnetism | Heaviside amplifier (PAPER\_1072) |

| 6 (Buoy) |  $F_{U,Bi}$  buoyancy | Variational EOM (PAPER\_1065) |  
 | 7 (Aether) | Vacuum background | Two-component  $\rho$  (PAPER\_1051) |  
 | 8 (LENR) | Nuclear transmutation | COP parametric (PAPER\_1081) |  
 | 9 (KK) | Kaluza-Klein 26D |  $S_{26}^{(3)}$  compactification (PAPER\_1080) |

## Calibration Constants

| Constant               | Symbol                | Value                                 | Validation Domain             |
|------------------------|-----------------------|---------------------------------------|-------------------------------|
| UQFF damping rate      | $\kappa$              | $5.0 \times 10^{-4} \text{ day}^{-1}$ | Magnetar spin-down, GW events |
| String sector coupling | $[SSq]$               | 0.57                                  | BH dynamics, nuclear binding  |
| Buoyancy coupling      | $\beta_i$             | 0.603                                 | Multi-system calibration      |
| SCm phonon frequency   | $\omega_{\text{SCm}}$ | $2\pi \times 1.25 \text{ THz}$        | Phonon resonance              |
| SCm completeness       | $H_{\text{SCm}}$      | 0.99                                  | Vacuum structure              |
| SCm vacuum density     | $\rho_{\text{SCm}}$   | $7.09 \times 10^{-37} \text{ J/m}^3$  | Fundamental constant          |

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## SM Anchor --- CVW v2.0.0

| Observable           | UQFF Prediction                                         | Status           |
|----------------------|---------------------------------------------------------|------------------|
| BCS gap symmetry     | $\Delta \propto \tanh(\Delta/2k_{BT}) \cdot S_{26}$     | Mapped to SCm    |
| Gap closure at $T_c$ | $\Delta(T_c) = 0$ consistent with BCS                   | Validated        |
| Phonon coupling      | $\omega_{\text{SCm}} = 1.25 \text{ THz}$ replaces Debye | Novel prediction |

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## §A Cosmogenesis-Linked Lagrangian (PAPER\_877 Symbolic Export)

§A.1 Sector Classification **\*\*Sector:\*\* Superconducting Gap (BCS-SCm)**

§A.2 Lagrangian Density  $\mathcal{L}_{\text{gap}} = -\frac{N(0)|\Delta|^2}{V_{\text{SCm}}} + N(0)\hbar\omega_{\text{SCm}}\ln\!\left(2\cosh\frac{\Delta}{2k_{BT}}\right)$

§A.3 Euler-Lagrange Equation of Motion  $\boxed{\frac{\delta S}{\delta \Delta} = 0}$  implies  $\Delta = \frac{\hbar\omega_{\text{SCm}}}{2}\tanh\!\left(\frac{\Delta}{2k_{BT}}\right) \cdot S_{26} \cdot \frac{F_{U,Bi}}{F_U}$

§A.4 Cosmogenesis Linkage Chain PAPER\_877 axioms  $\rho_{\text{SCm}} \rightarrow \omega_{\text{SCm}}$  phonon  $\rightarrow$  BCS gap  $\rightarrow$  Cooper pair binding  $\rightarrow$  superconducting condensate

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## §B VDS/DVP/BSH Deep Synthesis

**§B.1 Vacuum Density Series (VDS)**  $\text{VDS}(r) = \rho_{\text{SCm}} \cdot \exp\left(-\frac{r}{\lambda_{\text{SCm}}}\right) \cdot S_{26}([SSq])$

**§B.2 Dipole Vortex Primes (DVP) Prime lattice encoding:**  $p_n \in \{2, 3, 5, 7, 11, 13\}$  maps to magnetic/non-magnetic shell hierarchy.

**§B.3 Buoyancy Saturation Harmonics (BSH)**  $\text{BSH}(n) = \tanh(\beta_i \cdot n / 26) \cdot S_{26}$

#### §B.4 Production-Scale Consistency

| Metric                                         | Value              | Status    |
|------------------------------------------------|--------------------|-----------|
| VDS ratio $\rho_{\text{SCm}}/\rho_{\text{UA}}$ | 0.1                | Confirmed |
| DVP prime                                      | 2 (Cooper pair)    | Confirmed |
| BSH layers                                     | 26                 | Confirmed |
| $\kappa$ decay                                 | $5 \times 10^{-4}$ | Confirmed |
| $[SSq]$                                        | 0.57               | Confirmed |

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## Appendix: Session 225 Cross-References (PAPER\_1000--1081)

> Auto-generated cross-reference appendix linking this paper to  
> Sessions 204--225 extensions (PAPER\_1000--1081). Added by  
> update\_corpus\_crossrefs.py (Session 225, April 2026).

| Paper      | Title                                             |
|------------|---------------------------------------------------|
| PAPER_1022 | GW Phonon Strain SCm Modulation of $h(t)$         |
| PAPER_1037 | AGN Buoyancy Jet Calculator --- SCm Jet Launching |
| PAPER_1004 | QGP Vacuum Density with SCm S26 Phonon Coupling   |
| PAPER_1079 | Galaxy Cluster Cooling-Flow Buoyancy Suppression  |
| PAPER_1020 | Cosmic Ray Phonon Acceleration DSA Spectrum       |
| PAPER_1021 | Pulsar Timing Phonon TOA Residual                 |
| PAPER_1023 | Neutrino Oscillation Phonon PMNS Matrix SCm       |
| PAPER_1024 | Magnetar Giant Flare SCm Phonon Reservoir         |
| PAPER_1033 | Galactic Bar Resonance SCm Pattern Speed          |
| PAPER_1043 | $F_{U_{Bi}}$ Multi-System Buoyancy Curve Sweep    |
| PAPER_1072 | SCm Activation Function Phonon Threshold          |
| PAPER_1073 | SCm Phonon-Driven Inflation Vacuum Buoyancy       |
| PAPER_1003 | Spectral Ladder Merger 26-State Hierarchy         |
| PAPER_1065 | Buoyancy Lagrangian EOM Variational Derivation    |
| PAPER_1069 | VDS-DVP-BSH Hybrid Calculator Unified             |
| PAPER_1049 | Source10 GPU DPM Spectral Atlas ALMA Overlay      |

16 cross-reference(s) identified.

### Key References with arXiv/DOI Identifiers

1. Abbott et al. (LIGO Scientific and Virgo Collaborations, 2016). *Observation of Gravitational Waves from a Binary Black Hole Merger*. Phys. Rev. Lett. **116**, 061102 — arXiv:1602.03837 — doi:10.1103/PhysRevLett.116.061102

2. Murphy, D. (2026). *Unified Quantum Field Framework (UQFF): Star-Magic v5.x Whitepaper Series*. Star-Magic Repository — [github.com/Daniel8Murphy0007/Star-Magic](https://github.com/Daniel8Murphy0007/Star-Magic)
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